A Deduced Theorem for Goursat-Type Integrals with Cube Roots

Theorem 1 (Analog for Cube Roots)**.** *Let $R(t)$ be a cubic polynomial with distinct roots. An integral of the form*

$$I = \int \frac{\mathfrak{F}(t)}{R(t)^{1/3}} dt \quad or \quad I = \int \frac{\mathfrak{F}(t)}{R(t)^{2/3}} dt$$

is elementary if it belongs to the family of Goursat-type pseudo-elliptic integrals[cite: 1]. Such an integral can be reduced by either a single substitution or by splitting the rational part $\mathfrak{F}(t)$ into at most two components, where each resulting integral is reducible by a single substitution.

Proof. The proof is a deduction based on the assertions presented in the source text[cite: 1]. The argument follows the structure established for the square-root case, adapted to the specific rules for cube roots as described.

1. Detection of Reducibility Type

The text states that prescriptions analogous to those in Goursat's 1887 paper exist for the cube root case. These allow one to detect whether the integral can be solved with a single substitution or if the rational part, $\mathfrak{F}(t)$, must be split.

2. Case 1: No Splitting Required

If the detection method indicates no splitting is needed, the text asserts that the integral can be rationalized by a single substitution. This substitution transforms the original integral into an elementary form. The resulting elementary integral will be of the type:

$$\int R_1(u^3) du \quad \text{or} \quad \int u \cdot R_2(u^3) du$$

where R_1 and R_2 are rational functions. These are integrable using standard methods.

3. Case 2: Splitting Required

If the detection method indicates that splitting is necessary, the text states that $\mathfrak{F}(t)$ must be decomposed into two components:

$$\mathfrak{F}(t) = \mathfrak{F}_1(t) + \mathfrak{F}_2(t)$$

The text explicitly notes that "three components are never needed," which is a key distinction from the square root case involving the Klein four-group of involutions. The original integral is thus broken into two separate integrals:

$$I = \int \frac{\mathfrak{F}_1(t)}{R(t)^{n/3}} dt + \int \frac{\mathfrak{F}_2(t)}{R(t)^{n/3}} dt \quad (n = 1 \text{ or } 2)$$

Each of these two component integrals then falls under Case 1, where each succumbs to a single rationalizing substitution.

Conclusion

Since any integral of this type can be reduced to elementary forms—either directly (Case 1) or via a two-part decomposition where each part is reducible (Case 2)—the theorem is established as described by the source text. □